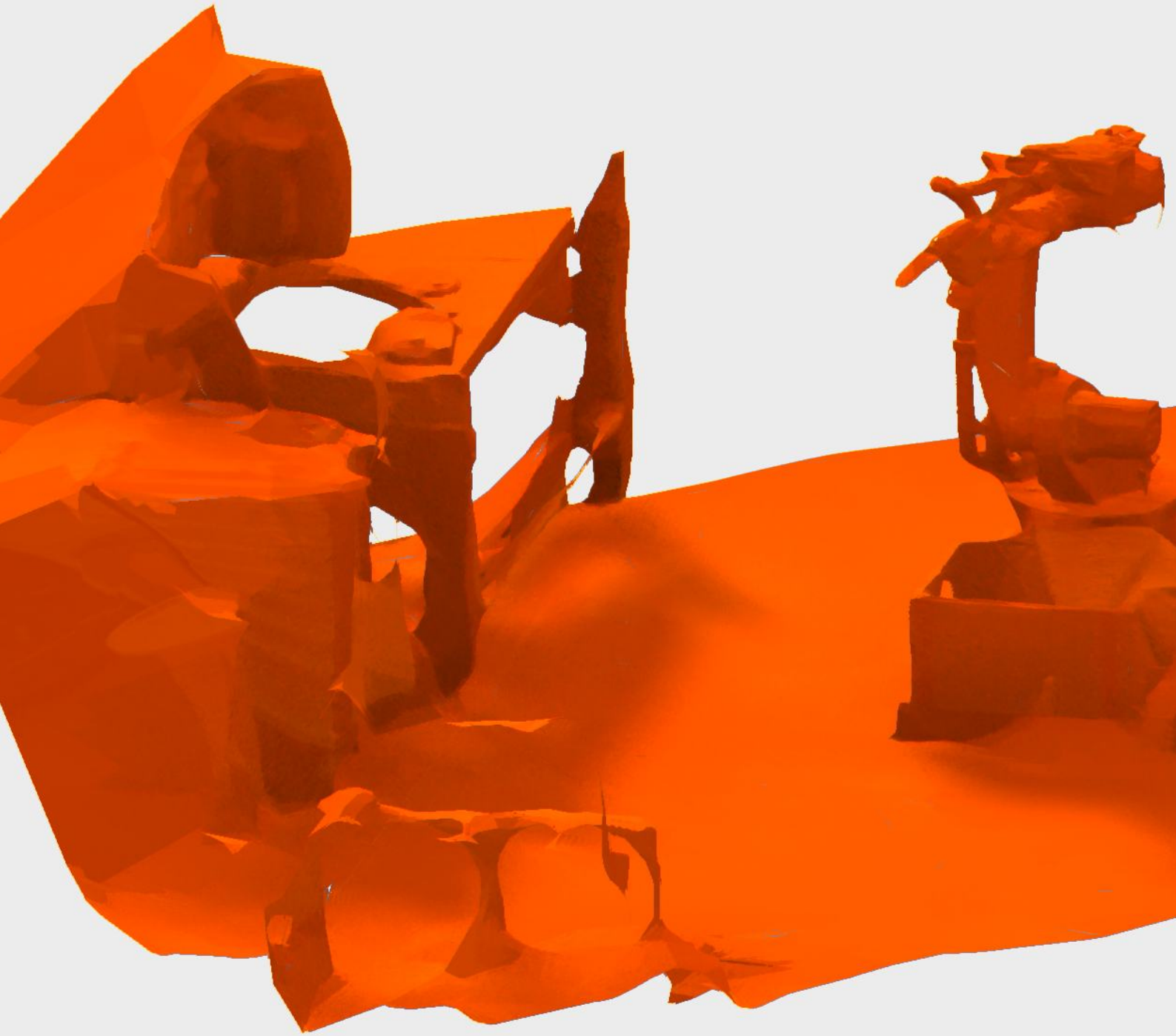
LAUDS Factories

# [DATA-RIDE] Blueprint



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# PROJECT INFORMATION

Mobility

This blueprint is used to document the problems addressed and the solutions developed within a project. It does not belong to the factory itself, but rather to a specific product created by the factory, encompassing everything related to that product or service.

**Project Description**

DATA-RIDE is a policy-driven citizen science project based on the open-source senseBox:bike platform, combining multiple data sources to support evidence-based advocacy and policy recommendations for safer urban cycling.

**Vision**

To make everyday cycling conditions visible, understandable and actionable by turning citizen-generated mobility data into evidence for safer, more inclusive and policy-driven urban cycling.

# KEY LEARNINGS

**KL01**

Collective interpretation creates impact
Workshops, public presentations and group discussions help transform raw data into shared understanding and useful advocacy messages.

**KL02**

Data collected through everyday rides shows cycling conditions that are often invisible in traditional planning processes because it's based on the real experiences

**KL03**

Cycling safety depends on interactions, not infrastructure alone
Safety emerges from the relationship between cyclists, drivers, pedestrians, street design, parking conditions and traffic behaviour.

# MAIN ISSUES

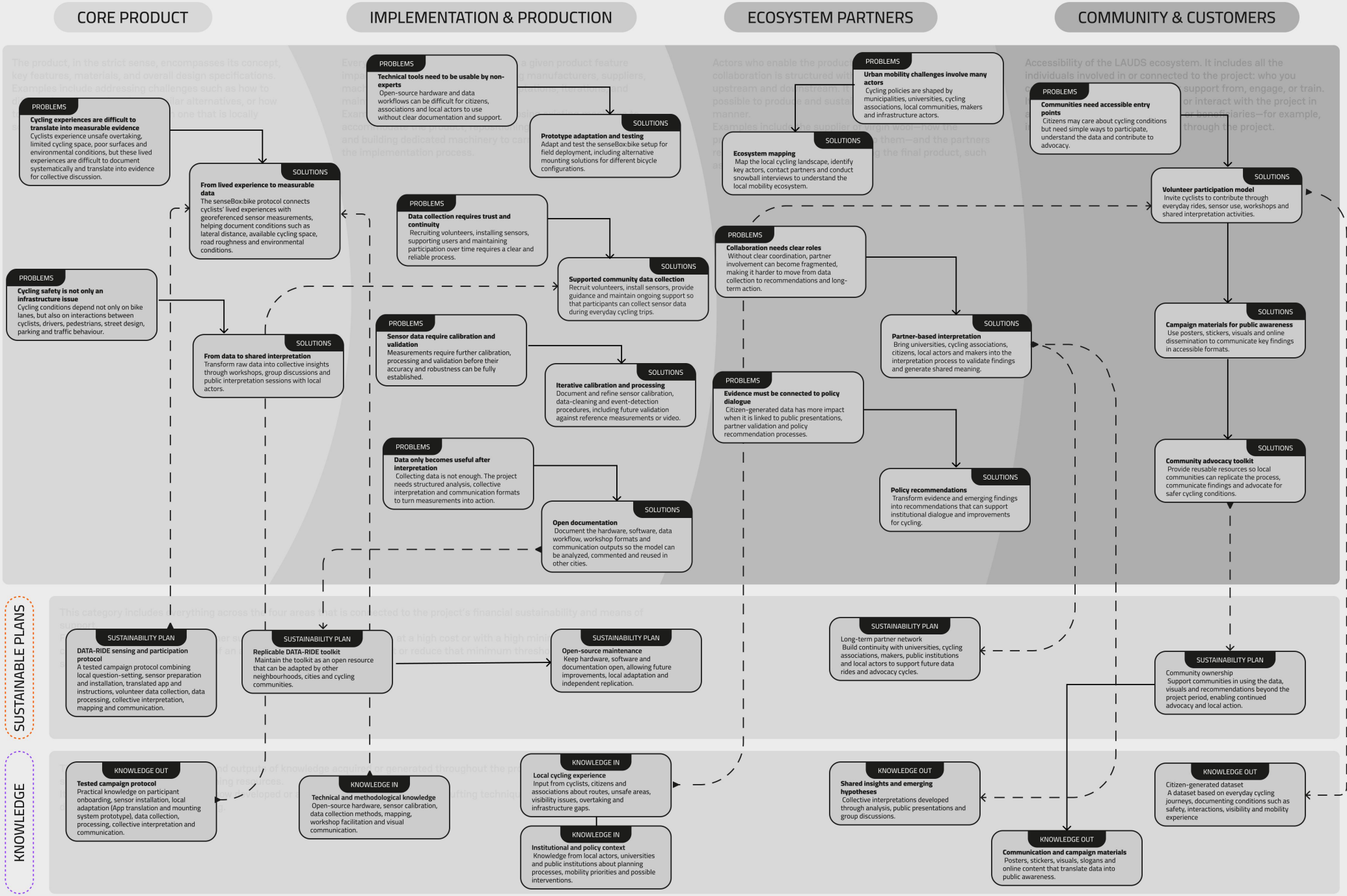
**MI01**

Data calibration and interpretation require more preparation time
Sensor calibration, data cleaning and collective interpretation require dedicated preparation time before findings can be translated into reliable insights.

**MI02**

Bike compatibility requires adaptable mounting solutions
Different bicycle models require different support designs, mounting strategies and testing procedures to ensure the sensor can be safely and reliably installed.

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CORE PRODUCT
IMPLEMENTATION & PRODUCTION
ECOSYSTEM PARTNERS
COMMUNITY & CUSTOMERS
PROBLEMS
Cycling experiences are difficult to translate into measurable evidence
Cyclists experience unsafe overtaking, limited cycling space, poor surfaces and environmental conditions, but these lived experiences are difficult to document systematically and translate into evidence for collective discussion.
SOLUTIONS
From lived experience to measurable data
The senseBox:bike protocol connects cyclists' lived experiences with georeferenced sensor measurements, helping document conditions such as lateral distance, available cycling space, road roughness and environmental conditions.
PROBLEMS
Cycling safety is not only an infrastructure issue
Cycling conditions depend not only on bike lanes, but also on interactions between cyclists, drivers, pedestrians, street design, parking and traffic behaviour.
SOLUTIONS
From data to shared interpretation
Transform raw data into collective insights through workshops, group discussions and public interpretation sessions with local actors.
PROBLEMS
Technical tools need to be usable by non-experts
Open-source hardware and data workflows can be difficult for citizens, associations and local actors to use without clear documentation and support.
SOLUTIONS
Prototype adaptation and testing
Adapt and test the senseBox:bike setup for field deployment, including alternative mounting solutions for different bicycle configurations.
PROBLEMS
Data collection requires trust and continuity
Recruiting volunteers, installing sensors, supporting users and maintaining participation over time requires a clear and reliable process.
SOLUTIONS
Supported community data collection
Recruit volunteers, install sensors, provide guidance and maintain ongoing support so that participants can collect sensor data during everyday cycling trips.
PROBLEMS
Sensor data require calibration and validation
Measurements require further calibration, processing and validation before their accuracy and robustness can be fully established.
SOLUTIONS
Iterative calibration and processing
Document and refine sensor calibration, data-cleaning and event-detection procedures, including future validation against reference measurements or video.
PROBLEMS
Data only becomes useful after interpretation
Collecting data is not enough. The project needs structured analysis, collective interpretation and communication formats to turn measurements into action.
SOLUTIONS
Open documentation
Document the hardware, software, data workflow, workshop formats and communication outputs so the model can be analyzed, commented and reused in other cities.
PROBLEMS
Urban mobility challenges involve many actors
Cycling policies are shaped by municipalities, universities, cycling associations, local communities, makers and infrastructure actors.
SOLUTIONS
Ecosystem mapping
Map the local cycling landscape, identify key actors, contact partners and conduct snowball interviews to understand the local mobility ecosystem.
PROBLEMS
Collaboration needs clear roles
Without clear coordination, partner involvement can become fragmented, making it harder to move from data collection to recommendations and long-term action.
SOLUTIONS
Partner-based interpretation
Bring universities, cycling associations, citizens, local actors and makers into the interpretation process to validate findings and generate shared meaning.
PROBLEMS
Evidence must be connected to policy dialogue
Citizen-generated data has more impact when it is linked to public presentations, partner validation and policy recommendation processes.
SOLUTIONS
Policy recommendations
Transform evidence and emerging findings into recommendations that can support institutional dialogue and improvements for cycling.
PROBLEMS
Communities need accessible entry points
Citizens may care about cycling conditions but need simple ways to participate, understand the data and contribute to advocacy.
SOLUTIONS
Volunteer participation model
Invite cyclists to contribute through everyday rides, sensor use, workshops and shared interpretation activities.
SOLUTIONS
Campaign materials for public awareness
Use posters, stickers, visuals and online dissemination to communicate key findings in accessible formats.
SOLUTIONS
Community advocacy toolkit
Provide reusable resources so local communities can replicate the process, communicate findings and advocate for safer cycling conditions.
SUSTAINABLE PLANS
SUSTAINABILITY PLAN
DATA-RIDE sensing and participation protocol
A tested campaign protocol combining local question-setting, sensor preparation and installation, translated app and instructions, volunteer data collection, data processing, collective interpretation, mapping and communication.
SUSTAINABILITY PLAN
Replicable DATA-RIDE toolkit
Maintain the toolkit as an open resource that can be adapted by other neighbourhoods, cities and cycling communities.
SUSTAINABILITY PLAN
Open-source maintenance
Keep hardware, software and documentation open, allowing future improvements, local adaptation and independent replication.
SUSTAINABILITY PLAN
Long-term partner network
Build continuity with universities, cycling associations, makers, public institutions and local actors to support future data rides and advocacy cycles.
SUSTAINABILITY PLAN
Community ownership
Support communities in using the data, visuals and recommendations beyond the project period, enabling continued advocacy and local action.
KNOWLEDGE
KNOWLEDGE OUT
Tested campaign protocol
Practical knowledge on participant onboarding, sensor installation, local adaptation (App translation and mounting system prototype), data collection, processing, collective interpretation and communication.
KNOWLEDGE IN
Technical and methodological knowledge
Open-source hardware, sensor calibration, data collection methods, mapping, workshop facilitation and visual communication.
KNOWLEDGE IN
Local cycling experience
Input from cyclists, citizens and associations about routes, unsafe areas, visibility issues, overtaking and infrastructure gaps.
KNOWLEDGE IN
Institutional and policy context
Knowledge from local actors, universities and public institutions about planning processes, mobility priorities and possible interventions.
KNOWLEDGE OUT
Shared insights and emerging hypotheses
Collective interpretations developed through analysis, public presentations and group discussions.
KNOWLEDGE OUT
Communication and campaign materials
Posters, stickers, visuals, slogans and online content that translate data into public awareness.
KNOWLEDGE OUT
Citizen-generated dataset
A dataset based on everyday cycling journeys, documenting conditions such as safety, interactions, visibility and mobility experience

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